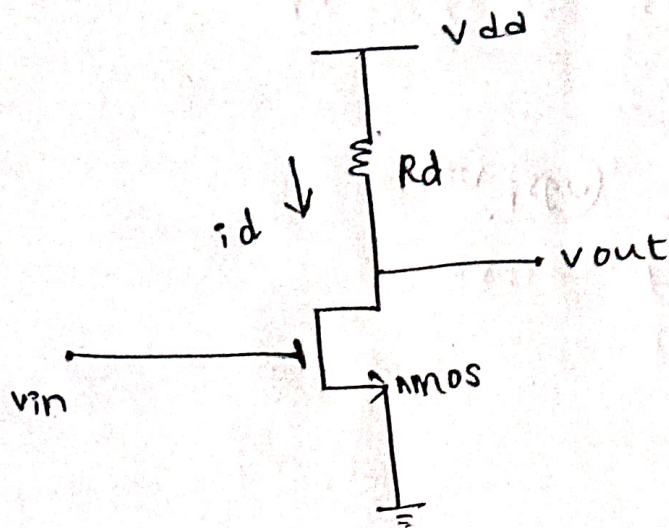


Common source Amplifier



$$V_{gs} = v_{in}$$

$$V_{ds} = v_{out}$$

$$V_{DD} - i_d R_d = v_{out}$$

for amplification, saturation region.

$$* I_D \Rightarrow \mu_n C_{ox} \left(\frac{W}{L} \right) \frac{(V_{gs} - V_{th})^2}{2}$$

$$v_{out} = V_{DD} - i_d R_d$$

$$v_{out} \Rightarrow V_{DD} - \left[\mu_n C_{ox} \left(\frac{W}{L} \right) \frac{(V_{gs} - V_{th})^2}{2} \right] R_d$$

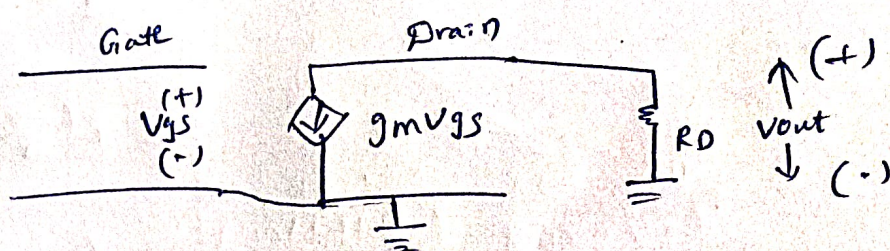
$$g_m = \frac{\partial I_D}{\partial V_{gs}} \Rightarrow \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{gs} - V_{th})$$

$$* \quad \frac{v_{out}}{v_{in}} = -g_m R_D$$

180° phase shift

Gain is proportional to ~~drain~~ resistance and
Transconductance.

from small signal model.



$$v_{out} = -g_m v_{gs} (R_D)$$

$$\frac{v_{out}}{v_{in}} = -g_m (R_D) \quad \text{without channel length modulation.} \\ (r_o = \infty)$$

$$\frac{v_{out}}{v_{in}} = -g_m (r_o \parallel R_D) \quad \text{with channel length modulation.}$$